

Laws of Indices — Paper C

HARD • NON-CALCULATOR • 45 MINUTES • 30 MARKS

NAME

DATE

SCORE / 30

- Answer **every** question in the space provided. Show your working — method marks are awarded even when the final answer is wrong.
- No calculator. Every number on this paper is small enough to handle on paper, and every fraction comes out simple.
- The laws are not printed on this paper. You are expected to know them.
- The mark for each question is shown in square brackets on the right.
- Several questions describe steps you may not have seen before — apply the ideas you already have. If two bases do not match, ask whether one of them can be rewritten in terms of the other.
- If a question stops you, leave it and come back — do not spend more than five minutes stuck.

SECTION 1 – NEGATIVE INDICES

1. Write each of these as an ordinary fraction.

(a) 2^{-4} [1]

(b) 5^{-2} [1]

(c) 10^{-3} [1]

Every answer here is a small positive number. A negative index is an instruction to flip, not a sign.

2. Write each of these as a power with a negative index.

(a) $1/9$, as a power of 3 [1]

(b) $1/32$, as a power of 2 [1]

(c) $1/100$, as a power of 10

[1]

3. Consider $2^3 \div 2^7$.

(a) Use the division law to write it as a single power of 2, and then as a fraction.

[2]

(b) Write both powers out in full and cancel, to show that your answer is right.

[1]

SECTION 2 – WHEN THE BASES DO NOT MATCH

4. Powers of 2 are hiding inside other numbers.

(a) Write 4 as a power of 2.

[1]

(b) Write 8 as a power of 2.

[1]

(c) Hence write $2^5 \times 4^2$ as a single power of 2.

[1]

5. Now the same idea with base 3.

(a) Write 9^3 as a power of 3.

[1]

(b) Hence write $9^3 \times 3^2$ as a single power of 3. [1]

(c) Write $27 \times 9^2 \div 3^5$ as a single power of 3, and work out its value. [2]

Convert every term to the same base before you reach for a law. Look for this whenever you see 4, 8, 9, 16, 25 or 27.

6. Simplify fully.

(a) $(2a^3)^2 \times 3a^4$ [2]

(b) $24x^7 \div (2x^2)^3$ [2]

SECTION 3 – MULTI-STEP

7. Write $16^2 \div 2^5$ as a single power of 2, and work out its value. [3]

8. Work out the value of each of these, giving your answer as a whole number or a fraction.

(a) $2^{-3} \times 2^5$ [2]

(b) $3^4 \times 3^{-6}$

[2]

The laws work exactly the same way when an index is negative. Add the indices first, then decide what the result means.

9. Write these four values in order, smallest first. Show the value of each one. [3]
- 2^{-1} 3^0 2^{-3} 3^1

End of paper. Check every answer has a unit or a form the question actually asked for.